

Image 4: The Hindenburg airship crossing the Atlantic on 4 April 1936. George Lewis would cross on the airship several months later on his epic tour of German research laboratories. (National Archives LC–USZ62–70162)



Chapter 1

Premonition | The Need for an Engine Tunnel (1936–1940)

With only the cabin windows, engine nacelles, and running lights illuminated, the new German luxury airship hung surreally in the darkened sky as Dr. George Lewis arrived at the field on the night of Monday, 21 September 1936. Dr. Lewis and the other passengers filed up the long gangway into the hovering ship. Upon reaching the top of the stairs leading to the main deck, Lewis came face to face with a bust of Field Marshal Paul von Hindenburg.[2] Shortly afterward, the *Hindenburg* left its mooring and silently carried Lewis away from Lakehurst, New Jersey, on a journey that would change the way the United States approached aeronautical research.

The 54-year-old aviation veteran was impressed by the massive airship's calm and steadiness, even later when traversing the wake of a passing hurricane. As Director of Aeronautical Research for the National Advisory Committee for Aeronautics, Lewis was responsible for coordinating the nation's aviation research. The *Hindenburg's* silent comfort during the 55-hour passage may have only increased Lewis's foreboding thoughts on what lay ahead of him across the ocean.

Lewis's journey to Germany and Russia was spurred by an invitation from the Deutsche Zeppelin-Reederei to tour the German aeronautical facilities. Despite the dominance of U.S. air manufacturers, Lewis was aware that the Europeans were ahead in the development of high-powered engines.[3] A letter from the NACA's European liaison, John Jay Ide, had already piqued Lewis's interest with its description of a large pressurized wind

tunnel at Göttingen, and visiting Russian researchers had claimed that their aeronautical research institute had 3,500 employees.[4] The NACA consisted of a single research laboratory, Langley Aeronautical Laboratory, with only 350 employees.

Both the economy and aircraft industry had boomed in the United States following World War I while Europe suffered. By the time of Lewis's trip in 1936, Boeing and Douglas possessed the premier airlines in the world. During this period Europeans were exploring ways to fly higher and faster, while Americans were easing up on their research and development. As a result, the United States had greater quantities of aircraft but would find itself lagging behind in regards to propulsion advances such as liquid-cooled and turbojet engines.[5]

"I Have Seen Nothing Like Them in America"

German Chancellor Adolf Hitler had created an air ministry, the Deutsche Versuchsanstalt für Luftfahrt, and supplied it with ample funding for extensive research and development. Lewis's personal guide in Berlin was



Image 5: Dr. George Lewis managed the aeronautical research conducted at the NACA for over 20 years. His most important accomplishment, however, may have been an investigative tour of research facilities in Germany in the fall of 1936. A second tour in 1939 included Germany, Britain, and France. The visits resulted in the NACA's physical expansion and the broadening of the scope of its research. (NASA EL–1997–00143)

Dr. Adolf Baeumker, the head of all German aeronautical research and development projects. When Baeumker told Lewis, "the ordinary military man is only interested in getting a large number of airplanes and not in aeronautical development and research," he was praising Air Minister Hermann Göring, but he could just as well have been referring to the field of aviation in their respective countries.[6]

Baeumker explained that Germany had developed a long-term aeronautical research plan that involved three major research laboratories. In what would become a model for the NACA, the Deutsche Versuchsanstalt für Luftfahrt had expanded its primary laboratory near Berlin and created two new research labs—one for aerodynamics and the other for engine research. Lewis estimated the total employee contingent in the German air ministry to be 1,600 to 2,000 and growing. Lewis remarked (about the German laboratories), "I have seen nothing like them in America." Lewis concluded, "If the United States is to hold its present position in the field of aeronautical research, it will be necessary to increase the personnel at Langley Field [Langley Aeronautical Laboratory]."[7]

Lewis was also shown a 22- by 17-foot pressure tunnel at Göttingen University. Göttingen had been the site of the first modern wind tunnel and had helped to forge the basis of the internally supported wing span. Lewis referred to the facility, which was capable of creating pressure levels three times that found at sea level while maintaining a useful Reynolds number, as the "first wind tunnel of this type to be constructed in the world."[8] Lewis later confided that he was likely the first foreigner to ever visit the Göttingen facility.[9]

Evolution of the Wind Tunnel

The concept of trying to re-create flight in Earth-bound facilities had developed during the 19th century in Europe. The Wright Brothers constructed a small wind tunnel in 1901 that yielded significant lift information for their early flights. However, it was Albert Zahm's tunnel at Catholic University, also constructed in 1901, which became the first significant U.S. tunnel. The 50-foot-long tunnel with its 6- by 6-foot test section dwarfed any of its contemporaries.[10] Zahm's tunnel and those following it benefited from replacing steam engines with more efficient electric-powered engines, which resulted in greater wind speeds at a lower cost.[11] Though Zahm's tunnel was hampered by occasional electrical variations and the effect of external weather conditions, the airflow controls and instrumentation would be used by other tunnels for years.[12]

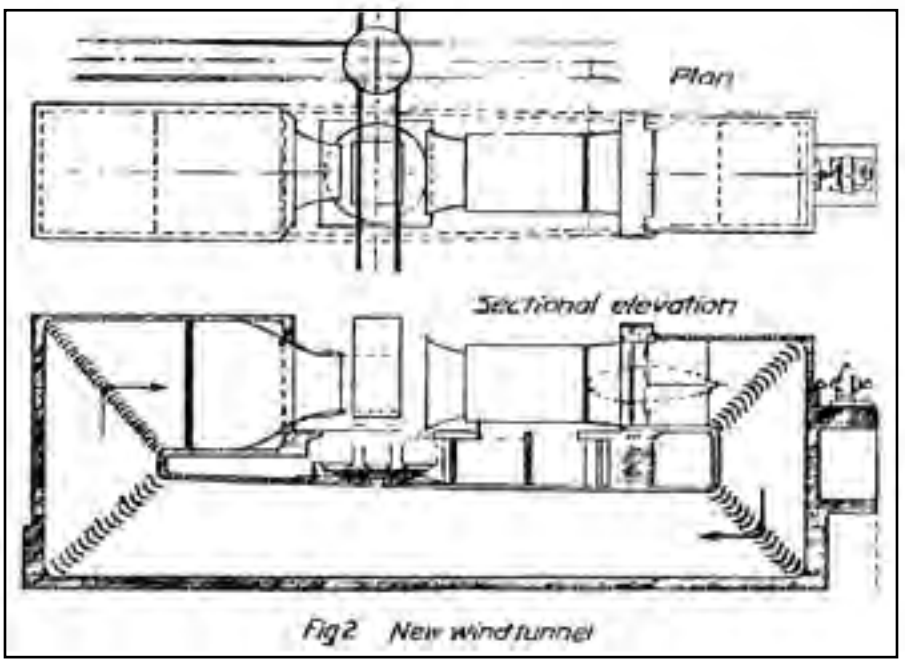


Image 6: Ludwig Prandtl's 1917 wind tunnel at Göttingen University in Germany. Prandtl's closed-loop tunnel revolutionized wind tunnel design. It included a throat section, turning vanes, and an airflow straightener. Studies in the Göttingen tunnel included wing profile examinations, full-scale propeller tests, and studies of the boundary layer on a rotating cylinder.[13] *(1920) (NACA TN No. 66, Fig. 2).*

Despite these early advances, the field of aeronautics in the United States was led primarily by nonprofessionals, whereas in Europe many scientists and engineers were pushed into the field by their governments. In addition, U.S. military leaders did not appreciate the importance of aircraft. The lack of research and government interest led to a deficiency of aircraft in the nation that pioneered aviation. An NACA report states, "When World War I erupted in 1914, it was reported that France had 1400 airplanes, Germany 1000, Russia 800, Great Britain 400, and the United States 23!"[14]

Russia, France, and Great Britain all constructed substantial wind tunnels around the turn of the century. The most influential European facility was Ludwig Prandtl's tunnel built at the University of Göttingen. With funding from the German Society for Airship Study, Prandtl had his initial tunnel operating by 1909. It was the first closed-loop tunnel and the first to use turning vanes or "deflectors" to guide the airflow around the corners. From the beginning, this facility was seen as a stepping stone to a bigger, more complex tunnel. The design of this new tunnel had begun in 1911. After being delayed by the war, the new facility became operational in the spring of 1917.[15]

Prandtl's new tunnel expanded upon the innovations of the first. Its 120-mile-per-hour (mph) speeds made it the most powerful wind tunnel of the period. The tunnel's diameter was expanded to 20 square meters upstream and then narrowed sharply to 4 square meters just before the test section to increase the airspeed. A honeycomb screen was installed across the width of the tunnel to straighten the airflow without slowing its speed. The basic components of Prandtl's tunnels were standard on almost all later tunnels, including the AWT.[16]

The NACA Starts Its Tunnel Collection

Although its formation in 1915 was too late to have much of an influence on World War I, the NACA's staff and mission were substantially expanded after the war. In 1917 the NACA established its own research lab at Langley Field in Hampton, Virginia. Unlike those found in industry, the NACA Langley Aeronautical Laboratory's facilities were to be used primarily to study new aircraft designs and tackle anticipated future aeronautical problems for the civil aviation industry. One of the principal reasons for creating the lab was to build wind tunnels. The first was a small insignificant tunnel that became operational in 1921. Coincidentally this tunnel, the Atmospheric Wind Tunnel, was also referred to as the "AWT."[17]

Langley's next tunnel, the Variable Density Tunnel, however, was a major advancement. It was proposed in 1921 by Dr. Max Munk, who had worked under Prandtl at Göttingen before emigrating to the United States. The Variable Density Tunnel was the first tunnel to forgo normal airflow for highly compressed air. It included a large steel tank in which the atmosphere could be pressurized and a wooden test section that negated the Reynolds number differential between tunnel tests and actual flight conditions. The tunnel, which became operational in 1923, was a highly useful tool and temporarily closed the gap between U.S. and European research.[18]

Langley began putting an entire collection of increasingly complex wind tunnels into operation. The next was the Propeller Research Tunnel in 1927, followed by the Vertical Spin Tunnel and Atmospheric Wind Tunnel in 1930, the Full-Scale Tunnel in 1931, the 8-Foot High Speed Tunnel in 1936, and the 19-Foot Pressure Tunnel in 1939. The 8-Foot High Speed Tunnel, which could simulate pressure altitudes of 12,000 feet and speeds of 500 mph, and the 19-Foot Pressure Tunnel, which combined a large test section with 250-mph speeds, were significant steps forward in flight simulation.[19]



Image 7: Elton Miller, successor to Max Munk as Langley's chief of aerodynamics, stands in the exit cone of the Propeller Research Tunnel to view the Sperry M-1 Messenger during the tunnel's first runs. The Propeller Research Tunnel at Langley was used to clean up drag caused by aircraft engines but not to study the performance of those engines. The tests were run in atmospheric conditions and at relatively slow speeds. (1927) (NASA EL–01892)

Exile of the Engine

Early research at Langley dealt almost exclusively with aerodynamics, even in its propeller tunnel. The Propeller Research Tunnel could fit an entire aircraft with an operating engine in its 20-foot-diameter open test section and could test it at airspeeds up to 110 mph. The tunnel made considerable contributions to the aerodynamics of propellers and engine nacelles, including the "NACA Cowl," but little to the actual operation of the engines.[20] Langley's Powerplants Division, led by Carlton Kemper, had only a handful of people until 1938, when it increased to 12.[21] The group dealt with the fundamentals of engine power, efficiency, and fuel consumption, but it did not delve into specific problems associated with these elements.[22]

The NACA's lack of interest in propulsion seems to have stemmed from an early meeting that it held in 1916 with automobile and aviation engine manufacturers. The group agreed that with military funding for research, these manufacturers would possess the needed assets and skills to develop aircraft engines. Initially the NACA did not have facilities to test aircraft engines, so Samuel Stratton, who headed the National Bureau of Standards and served as an original NACA committee member, took on the testing for the manufacturers at the National Bureau of Standards laboratories. The NACA seemed content with this arrangement even after Langley was established. As late as 1937, an original NACA member, Dr. Joseph Ames, stated, "The [engine] problem is primarily and almost solely one of development, which can best be attacked by the aircraft engine industry under experimental contracts with the War and Navy Departments."[23]

In order to stretch limited propulsion funding and manpower, the Powerplants Division primarily studied single cylinders and then extrapolated the test data to full-scale engines. Abe Silverstein, who was working with the aerodynamicists in the Full-Scale Tunnel at Langley, discovered that each cylinder produced a unique temperature during engine operation. This phenomenon could not be studied using single cylinders. Individual engine components could operate well in isolation only to fail when integrated with the larger system. Langley was not equipped to test large full-scale engines, and Kemper continued to put the engine onus on the manufacturers.[24]

The coordinated national research program in Germany pursued engine technology as well as aerodynamics. On his 1936 tour, Lewis was also shown a Deutsche Versuchsanstalt für Luftfahrt facility that was able to test water- and air-cooled engines under simulated altitude conditions. The intake air and exhaust were kept at pressure and temperature levels that corresponded to altitude conditions. The extensive instrumentation and infrastructure led Lewis to wonder "whether one is in an engine-testing laboratory or a small edition of the Licke observatory."[25]

Construction of a larger facility to test engines would begin at Göttingen during World War II. It was a wind tunnel designed to study operating full-scale jet engine systems in simulated altitude conditions at speeds up to 290 mph. Its refrigeration system would produce temperatures of –64°F, allowing the tunnel to also be used for icing studies.[26] Although construction would only be 80 percent complete because of severe German defeats during the war, the tunnel would have been remarkably similar to the AWT in Cleveland.

Engine testing was taking place in the United States at this time, but it was scattered and uneven. Private companies could not afford to build large test

facilities. The National Bureau of Standards, the U.S. Naval Aircraft Factory, and the U.S. Army Air Corps at McCook Field had successfully designed tanks to test small engines that could simulate the temperatures and pressures associated with altitude, but they were of limited size and could not incorporate the benefits of a wind tunnel.[27] On the other hand, the 7- by 10-foot Wright Brothers Wind Tunnel at the Massachusetts Institute of Technology and the 19-Foot Pressure Tunnel at NASA Langley, both completed in 1939, could simulate altitudes in a wind tunnel setting but were not capable of running aircraft engines. The AWT and the unfinished Göttingen tunnel were intended to solve both sides of the equation.

Peace-Time War

George Lewis was not the only NACA member visiting Germany in the 1930s. Charles Lindbergh was living abroad to escape the publicity surrounding the kidnapping and murder of his son. From this vantage point, he was able to provide unique insight into the European aeronautical industry. In 1936 Lindbergh made the first of five trips to inspect German aeronautics facilities. He regularly wrote to Dr. Ames with his findings.

In a letter dated 28 November 1938, Lindbergh wrote that European military aircraft were willing to trade range for ever-increasing speed and altitude. He felt that the United States should concentrate its efforts on improving its aircraft designs instead of building additional aircraft. He concluded, "It is more necessary than ever before to give full scope to research if we are to regain the leadership we have lost in some of the fields of military aviation."[28] Ames responded the following week that General Oscar Westover, then Chief of the U.S. Army Air Corps, had initiated the NACA's new Future Research Facilities Special Committee to investigate the possibility of starting a new NACA laboratory. Ames, added, "There is a new atmosphere in Washington. It has been likened to a state of 'peace-time war.'"[29]

In response to General Westover's special committee report and the impending war, the military finally turned to the NACA for assistance. Congress approved funding for the expansion of the NACA in 1939 but felt that a new laboratory was needed to decongest the Langley research community and disperse the facilities to prevent a catastrophic enemy attack.[30] Initially the NACA sought a single additional facility in which to investigate high-speed flight. Lindbergh returned to the United States to lead a special committee to research sites for the new lab. The group selected Moffett Field in Sunnyvale, California, and construction of the NACA Ames Aeronautical Laboratory began on 20 December 1939.[31]

The Lindbergh committee report also stated, "There is a serious lack of engine research facilities in the United States, and that it is of the utmost importance for the development of aviation in general, and for our defense program in particular, to take immediate steps to remedy this deficiency."[32] It was evident that the NACA also required a laboratory to specifically study aircraft engines.

On 19 October 1939 the NACA endorsed the new research facility for aircraft propulsion systems. George Mead, former vice president of both Pratt & Whitney and United Aircraft Corporation, was selected to lead a Special Committee on New Engine Research Facilities. The team included George Lewis, Carlton Kemper, the U.S. Army's Major Edwin Page, and others. A committee report issued on 23 January 1940 called for a $10-million laboratory that would include a test stand for engines, a fuels and lubricants facility, and—after some debate—a wind tunnel for engines.[33,34]



Image 8: Left to right: Charles Lindbergh, U.S. Air Force Vice Admiral Arthur Cook, U.S. Navy Charles Abbot, and Dr. Joseph Ames. From Europe, Lindbergh wrote to fellow NACA member, Dr. Ames, "I believe we should accept the fact that Germany will continue to be the leading country in Europe in aviation. She will be the leading country in the world if we do not increase our own rate of development. Even now Germany is far ahead of us in military aviation." Lindbergh claimed that Germany's ever-growing research facilities were so new that it would be years before their impact could be determined.[35] (NASA GPN–2002–000024)



Image 9: General Henry Arnold greets fellow NACA committee member Dr. George Lewis. Arnold served on the main committee from 1938 to 1944 and was a strong advocate for a new engine research facility. Arnold believed in continual research and development. Some of his most influential advisors were civilians such as Theodore von Kármán, Robert Millikan, and Charles Lindbergh.[36] *(NASA C-1944-07495)*

NACA Executive Secretary John Victory remembered seeing a newspaper headline stating "German Army Enters Paris" while sitting in the Senate gallery during the June 1940 appropriation hearings. Although the Bureau of the Budget lowered the requested amount to $8.4 million, the figure was eventually increased to $13.3 million to accommodate a larger role for what was to become the AWT.[37]

Engine Research Tunnel

During the Alessandro Volta Foundation's 1935 conference on High Velocities in Aviation, Jakob Ackeret predicted the use of the gas turbine for high-speed, high-altitude aircraft propulsion. He also stated that new wind tunnels capable of simulating altitude were important since high-speed flights would likely take place at high altitudes. Ackeret envisioned a tunnel in which a vacuum or pressures up to 10 atmospheres could be maintained.[38]

Although the NACA participated in the conference, it was not until the impending war in 1939 that the importance of increasing altitudes for U.S. aircraft was thrust to the forefront of the NACA's consciousness. Improved antiaircraft weaponry would force bombers and fighters to fly higher in order to be effective. Air battles in World War I were generally fought at 100 mph and at an altitude around 10,000 feet. By early 1941, newspapers were reporting that World War II aircraft were already flying 400 mph at 20,000 feet. It was anticipated at the time that aircraft would be exceeding 500 mph and 30,000 feet in the near future.[39]

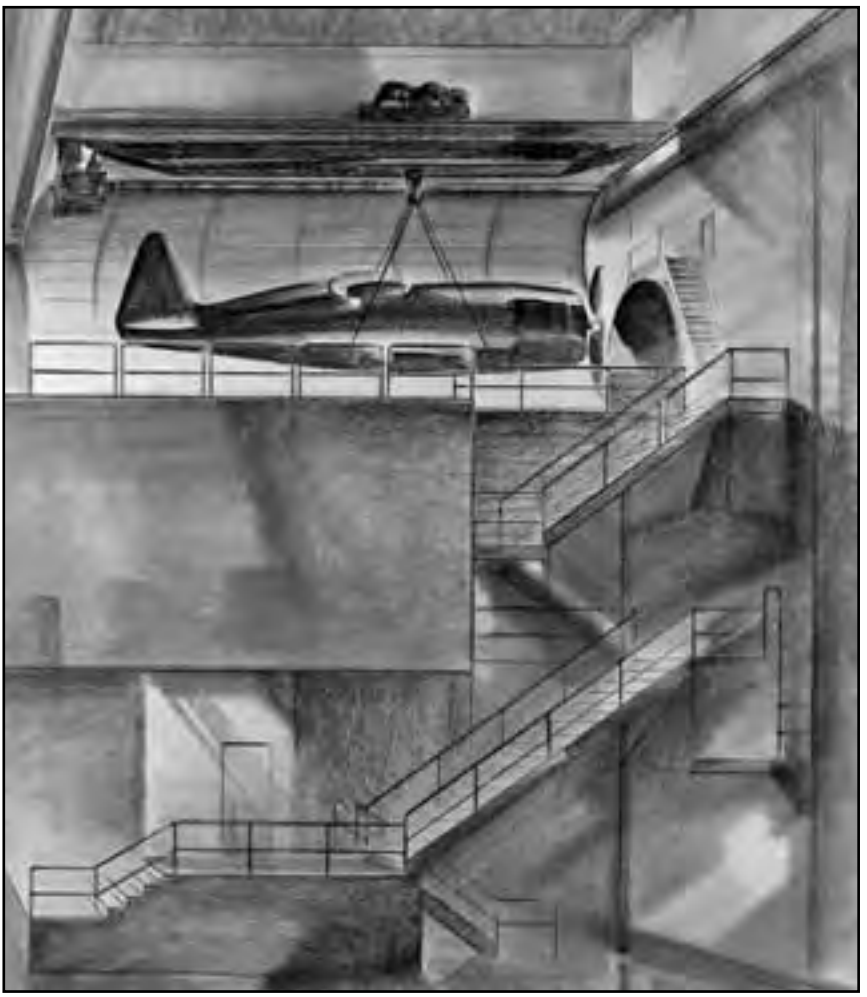


Image 10: A full-scale aircraft is lifted into the AWT's test section. Although testing full-size versions reduced the cost, time, and effort of developing engines, it also required a substantial investment in large facilities. Many of the NACA's achievements resulted from its willingness to make those investments. (NASA C-1944-06305)



Image 11: Early drawing of the new AWT, which was originally referred to as the "Engine Research Tunnel." It was the nation's first wind tunnel built to study engines under simulated flight conditions. (NASA C-1944-05308).

George Mead's Special Committee on New Engine Research Facilities rapidly began developing ideas and designs for test facilities at the new engine lab. Rudolph Gagg, a design consultant from Wright Aeronautical, said at the time, "This aviation research laboratory is being designed for the engines just around the corner. The purpose of the laboratory is to look forward to the engines of the future."[40] Although this would prove not to be true, the facilities would be robust enough to adapt to the new engines that would be around the corner.

The ability to test power, speed, drag, vibration, and cooling on complete engine systems would provide a faster transition from design to flight testing while avoiding additional time-consuming flight tests with risky and unproven engines. Performance in the harsh temperature and pressure conditions of these higher altitudes was vital.[41] The design and construction of the Engine Research Tunnel would prove to be a monumental task. It would combine elements from NACA Langley's Propeller Research Tunnel, 8-Foot High Speed Tunnel, and 19-Foot Pressure Tunnel with a massive refrigeration system and the ability to run large engines.

Not everyone agreed with the concept. Just weeks after the engine lab was approved, Mead wrote George Lewis, "I very much doubt whether there is a real need for the altitude chamber. In my opinion, a much more useful piece of equipment would be a full-scale tunnel in which we could test engines up to 4000-horsepower at sea-level pressure, the purpose being to check functional characteristics of multicylinder engines and their installations at speeds up to 500 miles per hour."[42]

Others saw the altitude tunnel as a vital research tool. Just days before the Pearl Harbor attack, William Knudsen, Director General of the War Department's Production Management Office, wrote to the NACA Chairman, Jerome Hunsaker, "The development of airplane engines of greater performance at high altitudes is absolutely essential." Referring to the AWT, Knudsen continued, "The high-altitude wind tunnel is especially needed to solve problems in connection with the cooling and power output of engines in combat planes required to fight at altitudes of 40,000 to 50,000 feet."[43]

Carlton Kemper said at the time, "AERL [the new engine lab—the Aircraft Engine Research Laboratory] is unique in having the only altitude wind tunnel in the world. We can expect that this one research tool will give answers to the military services that will more than offset the cost of the laboratory."[44] In the end, Mead would get much of what he sought, in addition to the altitude capabilities proposed by others.



Image 12: Zella Morewitz poses with a model of the NACA Aircraft Engine Research Laboratory, currently the NASA Glenn Research Center. The AWT can be seen near the center of the model. The AWT would be the research and geographical center of the lab for years. Morewitz transferred from Langley with the construction team and remained Dr. Ray Sharp's assistant for a number of years after the engine lab became operational. (NASA C-1942-01009)

Hello, Cleveland

The city of Cleveland put together a detailed proposal to persuade the NACA to build its engine lab in Cleveland. Industrial Commissioner Clifford Gildersleeve and Frederick Crawford, head of both the Chamber of Commerce and Thompson Products, Inc., were the city's principal advocates. Thompson Products manufactured aircraft parts locally, and Crawford himself organized and ran the National Air Races held annually since 1930 at the proposed site for the lab.[45] A crowd of 100,000 had attended the races over the 1939 Labor Day weekend.[46]

The NACA considered 72 bids in 62 other cities before announcing on 15 November 1940 that it had selected the 200-acre area north of the Cleveland Municipal Airport. The property had been used for parking and grandstands for the air races.[47] Proximity to an airport and the ability to generate sufficient



Image 13: Clevelanders swarm to watch the annual National Air Races at the Cleveland Municipal Airport on 5 September 1938. The city offered many advantages for the NACA: Clevelanders had long had an avid appreciation of aviation; Thompson Products was one of many aircraft manufacturing companies in the area; the Case Institute of Technology offered a supply of engineering graduates; and the adjacent airport would allow access for test aircraft. In a little more than two years, construction of the NACA's engine research lab would begin at this site. (NASA C–1991–01875)

electrical power were essential elements. The airport was a given, but the electricity posed a problem. The final hurdle was removed when Crawford negotiated an agreement in which the NACA would operate the AWT and other facilities requiring heavier power loads during the night. In exchange, the Cleveland Electric Illuminating Company promised sufficient power at reasonable rates.[48] This practice continues to this day.

During a luncheon at the Hotel Cleveland immediately before the 23 January 1941 groundbreaking for the Cleveland lab, Crawford reminded Cleveland manufacturers of their promise to supply the new lab with "anything it needs, more promptly, more cheaply, more accurately and more satisfactorily than it can be produced anywhere else."[49] John Victory, who had become friends with Gildersleeve by this time, said, "There seems to be something in the spirit of the people in Cleveland that makes effective cooperation seem easy."[50]

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